

JAXFluxTubeGK: Physical Model and Numerical Scheme for a Differentiable Flux-Tube Stellarator Gyrokinetic Solver

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Abstract

This document specifies the first implementation target for the project: a differentiable, local, flux-tube, linear electrostatic gyrokinetic solver for stellarator geometry. The physics, normalization, signs, and benchmarks follow the published GKW model, the original GKW Fortran source, and the differentiable Gyaradax implementation. The numerical target follows `task.tex`: Fourier discretization in the perpendicular directions and spectral discretization along the magnetic field and velocity-space coordinates. GKW-style finite differences are retained as a fallback and parity backend. Text is intentionally brief; equations, array contracts, numerical operators, and tests are written in enough detail to guide implementation in `src/`.

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1 Scope and Reference Hierarchy

The first solver shall implement the collisionless linear electrostatic local gyrokinetic system in a stellarator flux tube. The baseline electron model is adiabatic electrons with kinetic ions. Kinetic electrons, nonlinear ExB dynamics, collisions, electromagnetic perturbations, rotation, and global radial effects are documented as extensions, not as the first implementation target.

When conventions are ambiguous, use the following order of authority. GKW and Gyaradax are external reference codes: this project follows their published physics and, where useful, their source conventions, but does not vendor their source in this repository.

1. The published GKW paper [1].
2. GKW’s own Fortran source and sample cases, obtained as an external, revision-pinned checkout (see `docs/dependencies.md`).
3. The Gyaradax JAX implementation [2], likewise an external, revision-pinned checkout.
4. DESC and stellarator optimization papers for Boozer geometry and design-loop coupling.

The implementation must expose a pure residual

$$\partial_t f = \mathcal{R}(f; \phi, \mathcal{G}, p), \quad \phi = \Phi(f; \mathcal{G}, p), \quad (1)$$

where f is the gyrocenter distribution perturbation, ϕ is the electrostatic potential, $\mathcal{G}$ is the flux-tube geometry, and p collects species, profile, grid, and solver parameters. The residual is the central object for time stepping, eigenvalue solves, and automatic differentiation.

The primary discretization is:

$$(x, y) \text{ Fourier} \quad + \quad z \text{ spectral} \quad + \quad (v_{\parallel}, \mu) \text{ spectral}. \quad (2)$$

Finite-difference stencils from GKW/Gyaradax are not the default target for new solver kernels. They are a documented fallback for source-code parity, debugging, and convergence comparisons.

1.1 GKW Source Crosswalk

The filenames below are GKW’s own external Fortran modules, referenced here purely to record which module’s physics and conventions each part of this project follows.

Reference file	Use in this project
<code>linart.f90</code>	Main program flow and run sequencing.
<code>normalise.F90</code>	Normalization, growth-rate/frequency conventions.
<code>geom.F90</code>	Geometry, metric elements, drift tensors, circular and general geometry.
<code>mode.F90</code>	Fourier mode grids, radial/binormal mode labels, twist-and-shift connectivity.
<code>grid.F90, velocitygrid.F90</code>	Legacy grid sizes, velocity-space nodes, and weights for parity/fallback tests.
<code>components.F90</code>	Species parameters, Maxwellians, charges, masses, density and temperature ratios.
<code>functions.F90, specfun.f90</code>	Special functions used by Bessel and polarization factors.

<code>linear_terms.F90</code>	Linear gyrokinetic terms, Poisson/quasineutrality pieces, sign conventions.
<code>non_linear_terms.F90</code>	Future nonlinear ExB bracket, FFT/dealiasing, nonlinear timestep estimate.
<code>exp_integration.F90</code>	Explicit RK schemes, RHS assembly, field update flow.
<code>matdat.F90</code>	Matrix preparation, fallback stencil coefficients, linear timestep estimates.
<code>diagnostic.F90</code>	Growth rates, frequencies, fluxes, spectra, and output normalization.
<code>collisionop.F90,</code> <code>rotation.F90</code>	Future extensions beyond the first collisionless non-rotating baseline.

2 Normalization

All implemented equations should use a dimensionless normalization close to GKW/Gyaradax. Let R_{ref} , B_{ref} , m_{ref} , T_{ref} , n_{ref} , and $Z_{\text{ref}}e$ be reference values. Define

$$v_{\text{th,ref}} = \sqrt{\frac{2T_{\text{ref}}}{m_{\text{ref}}}}, \quad t_{\text{ref}} = \frac{R_{\text{ref}}}{v_{\text{th,ref}}}, \quad \rho_{\text{ref}} = \frac{m_{\text{ref}}v_{\text{th,ref}}}{Z_{\text{ref}}eB_{\text{ref}}}. \quad (3)$$

The dimensionless variables are

$$\hat{t} = \frac{t}{t_{\text{ref}}}, \quad \hat{\mathbf{x}} = \frac{\mathbf{x}}{R_{\text{ref}}}, \quad \hat{B} = \frac{B}{B_{\text{ref}}}, \quad \hat{\phi} = \frac{Z_{\text{ref}}e\phi}{T_{\text{ref}}}, \quad \hat{f}_s = \frac{f_s}{n_{\text{ref}}/v_{\text{th,ref}}^3}. \quad (4)$$

The hats are dropped below. Species parameters are stored as

$$\hat{m}_s = \frac{m_s}{m_{\text{ref}}}, \quad \hat{Z}_s = \frac{Z_s}{Z_{\text{ref}}}, \quad \hat{T}_s = \frac{T_s}{T_{\text{ref}}}, \quad \hat{n}_s = \frac{n_s}{n_{\text{ref}}}, \quad v_{\text{th},s} = \sqrt{\frac{\hat{T}_s}{\hat{m}_s}}. \quad (5)$$

The local profile gradients are dimensionless inverse scale lengths

$$\omega_{n,s} = \frac{R_{\text{ref}}}{L_{n,s}} = -R_{\text{ref}} \frac{d \ln n_s}{dr}, \quad \omega_{T,s} = \frac{R_{\text{ref}}}{L_{T,s}} = -R_{\text{ref}} \frac{d \ln T_s}{dr}. \quad (6)$$

Implementation names should be close to Gyaradax/GKW: $\omega_{n,s} \leftrightarrow \text{rln}$, $\omega_{T,s} \leftrightarrow \text{rlt}$, $\hat{m}_s \leftrightarrow \text{mas}$, $\hat{Z}_s \leftrightarrow \text{signz}$, $\hat{T}_s \leftrightarrow \text{tmp}$, and $\hat{n}_s \leftrightarrow \text{de}$.

3 Coordinates and Phase Space

3.1 Flux-tube coordinates

Use a local field-aligned coordinate system

$$(x, y, z) \equiv (\text{radial, binormal, parallel}), \quad \mathbf{B} = \nabla\psi \times \nabla\alpha, \quad (7)$$

where ψ labels the flux surface and α labels a field line. In Boozer angles (θ, φ) , a standard choice is

$$\alpha = \theta - \iota(\psi)\varphi - \alpha_0, \quad \iota = \frac{1}{q}. \quad (8)$$

The coordinate z parameterizes distance along the selected field line. For compatibility with GKW/Gyaradax, a cell-centered grid on $z \in [-n_{\text{period}} + 1/2, n_{\text{period}} - 1/2]$ is acceptable for

analytic tests. For stellarators, z may be Boozer toroidal angle, Boozer poloidal angle, or an arc-length-like coordinate, provided the geometry supplies $\mathbf{b} \cdot \nabla z$.

The Fourier representation in the perpendicular plane is

$$h(x, y, z, v_{\parallel}, \mu, t) = \sum_{k_x, k_y} \hat{h}_{k_x, k_y}(z, v_{\parallel}, \mu, t) \exp[i(k_x x + k_y y)], \quad (9)$$

with real-field Hermitian symmetry represented by a nonnegative k_y half spectrum.

3.2 Phase-space state

The evolved variable for species s is

$$f_s = f_s(v_{\parallel}, \mu, z, k_x, k_y, t), \quad (10)$$

where $\mu = v_{\perp}^2/(2B)$ in physical variables and is represented on a positive grid. The first implementation may use one kinetic ion species with adiabatic electrons:

$$f \in \mathbb{C}^{N_v \times N_{\mu} \times N_z \times N_x \times N_y}. \quad (11)$$

The multi-species extension adds a leading species axis:

$$f \in \mathbb{C}^{N_s^{\text{sp}} \times N_v \times N_{\mu} \times N_z \times N_x \times N_y}. \quad (12)$$

4 Geometry Required by the Solver

At every z_j , the solver needs a geometry object $\mathcal{G}(z_j)$ that supplies enough information to form $k_{\perp}^2$, parallel streaming, mirror force, ExB response, and magnetic drifts. The minimal internal geometry contract is

$$\mathcal{G} = \{B, F, G, E_y, D_x, D_y, g^{xx}, g^{xy}, g^{yy}, w_z, \text{mode connectivity}\}. \quad (13)$$

This maps naturally to Gyaradax/GKW-style arrays:

This document	Gyaradax/GKW-like name
$B(z)$	bn
$F(z)$, parallel streaming coefficient	ffun
$G(z) = -B^{-1}\partial_z B$ up to coordinate factors	gfun
$E_y(z)$, ExB/profile-drive coefficient	efun
$(D_x, D_y)(z)$, magnetic drift coefficients	dfun[:,0:2]
$(g^{yy}, g^{xy}, g^{xx})(z)$	little_g[:,0:3]
w_z	ints

The perpendicular wavenumber is evaluated as

$$k_{\perp}^2(z; k_x, k_y) = k_x^2 g^{xx}(z) + 2k_x k_y g^{xy}(z) + k_y^2 g^{yy}(z). \quad (14)$$

Numerically, enforce $k_{\perp}^2 \geq 0$ up to a small tolerance. Negative values beyond roundoff indicate an invalid metric or coordinate conversion.

4.1 Stellarator geometry import

The first imported-geometry interface uses ρ as the default radial surface coordinate, interpreted as a minor-radius-like normalized label. The adapter may also carry metadata for ψ or x , but the internal solver contract uses the local radial coordinate x and stores the chosen source coordinate as metadata. Do not differentiate through changes of radial coordinate convention.

For stellarator flux tubes, compute or load the following physical quantities along a field line:

$$\mathcal{G}_{\text{phys}} = \{B, \mathbf{b} \cdot \nabla z, |\nabla \psi|^2, |\nabla \alpha|^2, \nabla \psi \cdot \nabla \alpha, (\mathbf{B} \times \nabla B) \cdot \nabla \psi, (\mathbf{B} \times \nabla B) \cdot \nabla \alpha, (\mathbf{b} \times \boldsymbol{\kappa}) \cdot \nabla \psi, (\mathbf{b} \times \boldsymbol{\kappa}) \cdot \nabla \alpha\}, \quad (15)$$

where $\boldsymbol{\kappa} = \mathbf{b} \cdot \nabla \mathbf{b}$. The implementation should include an adapter

$$\mathcal{G}_{\text{phys}} \longmapsto \{B, F, G, E_y, D_x, D_y, g^{xx}, g^{xy}, g^{yy}\} \quad (16)$$

with tests on analytic circular geometry before using DESC/Boozer equilibria. The first supported import path is precomputed arrays on a sampled Boozer field line. DESC, SIM-SOPT/Boozer objects, VMEC, or booz_xform files should be added as source adapters that produce the same precomputed physical-array contract.

For coupling to DESC, the intended interface is array-based rather than a refactor of DESC inside the gyrokinetic package. DESC should solve or optimize the MHD equilibrium and evaluate the flux-tube quantities in (15) on the selected nodes z_i :

$$\mathcal{A}_{\text{DESC}}: \{E_{\text{DESC}}, \rho, \alpha, z_i\}_{i=1}^{N_z} \longmapsto \{\mathcal{G}_{\text{phys}}(z_i)\}_{i=1}^{N_z}. \quad (17)$$

The gyrokinetic solver then maps these arrays with (16), builds the fixed-topology residual, and evaluates the objective. If DESC exposes JAX-compatible sensitivities of $\mathcal{G}_{\text{phys}}$ with respect to equilibrium parameters p , the design gradient is assembled by the chain rule

$$\frac{\partial \mathcal{J}}{\partial p} = \sum_i \frac{\partial \mathcal{J}}{\partial \mathcal{G}_{\text{phys}}(z_i)} \frac{\partial \mathcal{G}_{\text{phys}}(z_i)}{\partial p}. \quad (18)$$

This keeps file I/O, equilibrium solves, and any remeshing outside the time-advance kernels while preserving differentiation through continuous geometry arrays.

The direct DESC extraction path uses DESC's field-line grid construction in (ρ, α, ζ) coordinates. For a chosen surface ρ_0 , field-line label α_0 , rotational transform $\iota(\rho_0)$, and toroidal nodes $\zeta_i = z_i$, DESC supplies the Cartesian vector quantities

$$\mathbf{B}, \quad \nabla|B|, \quad \nabla \psi, \quad \nabla \alpha, \quad \mathbf{b}, \quad \boldsymbol{\kappa}, \quad B^\zeta = \mathbf{B} \cdot \nabla \zeta.$$

The array adapter then forms

$$\mathbf{b} \cdot \nabla z = B^\zeta / |B|, \quad g^{\psi\psi} = \nabla \psi \cdot \nabla \psi, \quad (19)$$

$$g^{\alpha\alpha} = \nabla \alpha \cdot \nabla \alpha, \quad g^{\psi\alpha} = \nabla \psi \cdot \nabla \alpha, \quad (20)$$

$$C_\psi^B = (\mathbf{B} \times \nabla|B|) \cdot \nabla \psi, \quad C_\alpha^B = (\mathbf{B} \times \nabla|B|) \cdot \nabla \alpha, \quad (21)$$

$$C_\psi^\kappa = (\mathbf{b} \times \boldsymbol{\kappa}) \cdot \nabla \psi, \quad C_\alpha^\kappa = (\mathbf{b} \times \boldsymbol{\kappa}) \cdot \nabla \alpha. \quad (22)$$

These are the physical arrays expected by (15). A CLI fixture generator in `scripts/` loads either a DESC example equilibrium or a DESC HDF5/pickle path and writes the sampled arrays to an `.npz` file.

The same internal geometry can be exported to the `eik`-style contract used for stellarator parity checks:

$$\mathcal{E}_{\text{eik}} = \{\theta, B, \nabla_\parallel, \text{gds2}, \text{gds21}, \text{gds22}, \text{gbdrift}, \text{gbdrift0}, \text{cvdrift}, \text{cvdrift0}\}. \quad (23)$$

The implemented export uses

$$B = \text{bmag}, \quad \nabla_{\parallel} = F, \quad (\text{gds2}, \text{gds21}, \text{gds22}) = (g^{yy}, g^{xy}, g^{xx}),$$

and preserves the summed drift coefficients through

$$\text{gbdrift} + \text{cvdrift} = D_y, \quad \text{gbdrift0} + \text{cvdrift0} = D_x.$$

The split between grad- B and curvature drift pieces is source-code specific and is not inferred by the exporter. The internal mirror-force coefficient G is therefore checked separately unless an external reference provides a compatible proxy. This convention lets DESC-generated flux-tube arrays pass through the same metric/drift and $k_{\perp}^2$ contract used by external **eik** fixtures without claiming a curvature split that was not provided. Some external **eik**/GIST text fixtures use an alternate drift-column ordering; when reading that variant, the columns are read in the order

$$\theta, B, \nabla_{\parallel}, \text{gds2}, \text{gds21}, \text{gds22}, \text{cvdrift}, \text{cvdrift0}, \text{gbdrift}, \text{gbdrift0},$$

and the solver compares both the individual metric fields and the summed drift fields used by the residual.

4.2 Magnetic drift coefficient

Use a geometry-factorized magnetic drift frequency

$$\omega_{d,s}(z, v_{\parallel}, \mu, k_x, k_y) = \frac{v_{\parallel}^2 + \mu B(z)}{Z_s} [k_x D_x(z) + k_y D_y(z)], \quad (24)$$

where all dimensional constants are absorbed into the normalization and the geometry coefficients. This is the first implementation form because it matches the Gyaradax/GKW coefficient structure and is easy to test. Later, if curvature and grad- B drifts are separated, retain (24) as the assembled operator interface.

5 Maxwellian, FLR Factors, and Polarization

The normalized background Maxwellian is

$$F_{M,s}(v_{\parallel}, \mu, z) = \frac{n_s}{(\pi T_s)^{3/2}} \exp \left[-\frac{v_{\parallel}^2 + 2\mu B(z)}{T_s} \right], \quad (25)$$

where T_s and n_s are normalized species values. Define the normalized particle energy and thermodynamic drive factor

$$\mathcal{E}_s(v_{\parallel}, \mu, z) = \frac{v_{\parallel}^2 + 2\mu B(z)}{T_s}, \quad \Xi_s = \omega_{n,s} + \omega_{T,s} \left(\mathcal{E}_s - \frac{3}{2} \right). \quad (26)$$

The gyroaverage of ϕ is algebraic in Fourier space:

$$\langle \phi \rangle_s = J_0(a_s) \phi, \quad a_s = \frac{m_s v_{\text{th},s}}{|Z_s|} \frac{k_{\perp}}{B} \sqrt{2\mu B}. \quad (27)$$

The polarization factor uses

$$b_s = \frac{1}{2} \left(\frac{m_s v_{\text{th},s}}{Z_s} \frac{k_{\perp}}{B} \right)^2, \quad \Gamma_0(b_s) = I_0(b_s) e^{-b_s}. \quad (28)$$

Implementation detail: use a numerically stable exponentially scaled Bessel function for Γ_0 , e.g. **i0e**, and test the small- b limit $\Gamma_0(b) = 1 - b + \mathcal{O}(b^2)$.

6 Linear Electrostatic Gyrokinetic Model

The first implementation evolves the collisionless linear electrostatic system

$$\partial_t f_s = \mathcal{R}_s(f_s, \phi) = \mathcal{T}_{I,s} + \mathcal{T}_{II,s} + \mathcal{T}_{IV,s} + \mathcal{T}_{V,s} + \mathcal{T}_{VII,s} + \mathcal{T}_{VIII,s} + \mathcal{D}_s. \quad (29)$$

Term numbers follow the GKW/Gyaradax convention. Term III, the nonlinear ExB term, is omitted in the linear baseline and documented in Appendix A.

Let

$$a_{\parallel,s}(z, v_{\parallel}) = v_{\text{th},s} F(z) v_{\parallel}, \quad a_{\mu,s}(z, \mu) = v_{\text{th},s} \mu B(z) G(z). \quad (30)$$

Then the six active linear terms are:

$$\mathcal{T}_{I,s} = -a_{\parallel,s} \partial_z f_s, \quad \text{parallel streaming,} \quad (31)$$

$$\mathcal{T}_{II,s} = -i \omega_{d,s} f_s, \quad \text{magnetic drift advection,} \quad (32)$$

$$\mathcal{T}_{IV,s} = a_{\mu,s} \partial_{v_{\parallel}} f_s, \quad \text{mirror force/trapping,} \quad (33)$$

$$\mathcal{T}_{V,s} = i k_y E_y(z) J_{0,s} \phi F_{M,s} \Xi_s, \quad \text{equilibrium-gradient drive,} \quad (34)$$

$$\mathcal{T}_{VII,s} = -\frac{Z_s}{T_s} a_{\parallel,s} F_{M,s} \partial_z (J_{0,s} \phi), \quad \text{parallel field drive,} \quad (35)$$

$$\mathcal{T}_{VIII,s} = -\frac{Z_s}{T_s} i \omega_{d,s} F_{M,s} J_{0,s} \phi, \quad \text{drift field drive.} \quad (36)$$

All products are pointwise over the discrete phase-space grid after broadcasting over species, velocity, field-line, and Fourier axes.

6.1 Dissipation

The baseline solver includes only numerical dissipation:

$$\mathcal{D}_s = \mathcal{D}_{z,s} + \mathcal{D}_{v_{\parallel},s} + \mathcal{D}_{\perp,s}. \quad (37)$$

For GKW/Gyaradax-style parallel recurrence control, keep the damping inside the residual rather than applying a post-step filter:

$$\mathcal{D}_{z,s} f_s = -\text{disp_par} |a_{\parallel,s}^{\text{rms}}(z)| \frac{\Delta z^3}{12} \partial_z^4 f_s. \quad (38)$$

On a uniform finite-difference fallback this is exactly the GKW stencil $[-1, 4, -6, 4, -1]/(12\Delta z)$; on the spectral backend it is the low-wavenumber-equivalent fourth derivative with the same resolution scaling. Additional modal filtering is treated only as an experimental diagnostic. In a coordinate $\xi \in \{z, v_{\parallel}, \mu\}$,

$$\mathcal{D}_{\xi,s} f_s = -\nu_{\xi} T_{\xi}^{-1} \text{diag} \left[\left(\frac{r}{r_{\text{max}}} \right)^{p_{\xi}} \right] T_{\xi} f_s, \quad r = 0, \dots, r_{\text{max}}, \quad (39)$$

where T_{ξ} is the Fourier or Chebyshev modal transform. Set $\nu_{\xi} = 0$ by default. GKW-style fourth-order damping is used only in the finite-difference fallback backend.

The perpendicular directions remain Fourier, so perpendicular hyper-dissipation is

$$\mathcal{D}_{\perp,s} f_s = - \left[\nu_x \left(\frac{|k_x|}{k_{x,\text{max}}} \right)^{p_x} + \nu_y \left(\frac{|k_y|}{k_{y,\text{max}}} \right)^{p_y} \right] f_s, \quad (40)$$

$$p_x, p_y = \begin{cases} 4, & \nu_x, \nu_y \geq 0, \\ 2, & \nu_x, \nu_y < 0 \end{cases} \quad (\text{GKW-style option}). \quad (41)$$

For the first implementation, set default $\nu_x = \nu_y = 0$ unless a benchmark specifies otherwise. This avoids accidental damping of Rosenbluth-Hinton and linear ITG benchmarks.

7 Quasineutrality

7.1 Adiabatic electrons

For ion species i with adiabatic electrons, solve quasineutrality independently at each (z, k_x, k_y) :

$$\phi = - \frac{\sum_i Z_i n_i \int dv_{\parallel} d\mu B J_{0,i} f_i}{\sum_i \frac{Z_i^2 n_i}{T_i} [\Gamma_{0,i} - 1] - \frac{n_e}{T_e}}. \quad (42)$$

The discrete numerator uses the velocity weights w_v, w_{μ} :

$$N_i(z, k_x, k_y) = Z_i n_i \sum_{\ell, m} w_{v,\ell} w_{\mu,m} B(z) J_{0,i}(\ell, m, z, k_x, k_y) f_i(\ell, m, z, k_x, k_y). \quad (43)$$

For the zonal mode $k_y = 0$, GKW/Gyaradax apply a flux-surface correction to the adiabatic response. The implementation must keep this as a separately tested path. A practical contract is:

$$\phi_{k_y=0} = \Phi_{\text{local}}(f) + \Phi_{\text{zf correction}}(f), \quad (44)$$

where $\Phi_{\text{zf correction}}$ is matched against `linear_terms.F90` routines `poisson_dia` and `poisson_zf` and against Gyaradax tests.

7.2 Kinetic-electron extension

For fully kinetic species, use

$$\phi = - \frac{\sum_s Z_s n_s \int dv_{\parallel} d\mu B J_{0,s} f_s}{\sum_s \frac{Z_s^2 n_s}{T_s} [\Gamma_{0,s} - 1]}. \quad (45)$$

The zero-denominator zonal/constant mode must be regularized explicitly and tested. This path is not required before the adiabatic baseline passes tests. The present multispecies implementation passes an algebraic TEM-favorable preflight covering charge neutrality, kinetic quasineutrality, finite coupled residuals, electron thermal streaming scaling, and the explicit CFL estimate. This is not yet TEM validation: a converged growth rate, electron-diamagnetic-direction frequency, and complex mode structure must still agree with a revision-pinned independent solver.

8 Discrete Grids and Operators

The target backend is spectral in all non-perpendicular coordinates. Production kernels should therefore be written in terms of precomputed spectral quadrature weights and differentiation matrices. A finite-difference backend may implement the same public interface for GKW parity and fallback convergence studies.

8.1 Velocity-space spectral collocation

The first spectral velocity backend should keep the finite velocity box used by GKW-style benchmarks, but replace uniform finite differences with Chebyshev–Lobatto collocation. Let

$$\xi_\ell = -\cos \frac{\pi \ell}{N_v - 1}, \quad v_{\parallel, \ell} = v_{\max} \xi_\ell, \quad \ell = 0, \dots, N_v - 1, \quad (46)$$

and

$$\eta_m = \frac{1 - \cos \frac{\pi m}{N_\mu - 1}}{2}, \quad \mu_m = \mu_{\max} \eta_m, \quad m = 0, \dots, N_\mu - 1. \quad (47)$$

Let W_N^C and D_N^C denote Clenshaw–Curtis quadrature weights and the Chebyshev–Lobatto first-derivative matrix on $[-1, 1]$. The physical weights and derivative matrices are

$$w_{v, \ell} = v_{\max} W_{N_v, \ell}^C, \quad w_{\mu, m} = \mu_{\max} W_{N_\mu, m}^{C, [0, 1]}, \quad D_{v_{\parallel}} = \frac{1}{v_{\max}} D_{N_v}^C, \quad D_\mu = \frac{1}{\mu_{\max}} D_{N_\mu}^{C, [0, 1]}. \quad (48)$$

Here $W^{C, [0, 1]}$ and $D^{C, [0, 1]}$ are the mapped weights and derivative matrix on $[0, 1]$. Velocity integrals use

$$\int_{-v_{\max}}^{v_{\max}} dv_{\parallel} \int_0^{\mu_{\max}} d\mu B(z) g(v_{\parallel}, \mu, z) \approx B(z) \sum_{\ell, m} w_{v, \ell} w_{\mu, m} g_{\ell m}(z). \quad (49)$$

The gyrophase factor is absorbed into the chosen normalization; if a benchmark requires an explicit 2π , it should enter the documented velocity measure, not an individual physics term. Velocity collocation is the selected production CPU representation: Chebyshev is the general reduced/differentiable path, GKW finite differences retain convention parity, and the midpoint/Gauss–Laguerre recipe is used by the source-matched W7-X gate. Hermite–Laguerre remains an experimental reduced discriminator until it is integrated with the complete stellarator residual and field solve and passes comparable convergence, parity, gradient, and timing gates.

8.2 Parallel spectral grid

The parallel coordinate z is the chosen field-line parameter. For a closed mode chain, use Fourier collocation along the concatenated connected coordinate:

$$z_j = z_0 + \frac{j L_z}{N_z}, \quad q_r = \frac{2\pi}{L_z} \begin{cases} r, & 0 \leq r \leq N_z/2, \\ r - N_z, & N_z/2 < r < N_z, \end{cases} \quad (50)$$

and

$$D_z = F_z^{-1} \text{diag}(iq_r) F_z, \quad w_{z, j} = L_z / N_z. \quad (51)$$

Here F_z is the discrete Fourier transform along the connected field-line chain. For an open or non-closing chain, use Chebyshev–Lobatto collocation on the finite interval and the corresponding derivative matrix. In both cases the streaming operator is $a_{\parallel, s} D_z f_s$; the geometry object supplies $\mathbf{b} \cdot \nabla z$, metric factors, and quadrature weights for the chosen coordinate.

For imported stellarator geometry, interpolate or evaluate B , metric terms, and drift coefficients on the same spectral z nodes. Do not hide the quadrature rule in the RHS; store z_j , $w_{z, j}$, D_z , and any modal transform needed by filters in the geometry/precompute object.

8.3 Fourier grids

The k_y grid is a nonnegative half spectrum:

$$k_{y,n} = n\Delta k_y, \quad n = 0, \dots, N_y - 1. \quad (52)$$

The k_x grid is centered:

$$k_{x,m} = (m - m_0)\Delta k_x, \quad m = 0, \dots, N_x - 1, \quad (53)$$

where m_0 indexes $k_x = 0$. In single-mode linear runs, $N_x = 1$ and $k_x = 0$ is allowed.

Parseval weights for the k_y half spectrum are

$$P(k_y) = \begin{cases} 1, & k_y = 0, \\ 2, & k_y > 0. \end{cases} \quad (54)$$

8.4 Twist-and-shift connectivity

At a parallel boundary, nonzonal modes connect to a shifted radial wavenumber. Represent this with integer maps

$$\text{ixplus}(m, n), \quad \text{ixminus}(m, n), \quad (55)$$

where invalid entries denote an open end of a connected mode chain. For axisymmetric tests, the shift should reproduce GKW mode-box behavior. For stellarators, compute the shift from the field-line twist:

$$k_x^+ = k_x + k_y \Delta_{\text{twist}}, \quad \Delta_{\text{twist}} = \left. \frac{\partial \alpha_{\text{end}}}{\partial x} \right|_{\psi_0} - \left. \frac{\partial \alpha_{\text{start}}}{\partial x} \right|_{\psi_0}, \quad (56)$$

then choose Δk_x so the shifted mode lands on an integer grid index when possible. The continuous twist can be differentiable, but the integer connectivity maps are static topology and must not be differentiated.

8.5 Spectral derivative contract

All derivative-using RHS terms should call backend operators:

$$\partial_z f \mapsto D_z f, \quad \partial_{v_{\parallel}} f \mapsto D_{v_{\parallel}} f, \quad \partial_{\mu} f \mapsto D_{\mu} f. \quad (57)$$

The implementation may store nodal-to-modal transforms T_{ξ} for filtering and diagnostics:

$$\hat{f}^{(\xi)} = T_{\xi} f, \quad f = T_{\xi}^{-1} \hat{f}^{(\xi)}. \quad (58)$$

For smooth manufactured functions, derivative tests should demonstrate spectral convergence until floating-point or interpolation error dominates. On nonsmooth or open-boundary tests, the code should still converge monotonically and the finite-difference backend provides a useful sanity check.

8.6 GKW finite-difference fallback

The following GKW/Gyaradax-style formulas are fallback operators, not the primary target discretization from `task.tex`. They are useful for parity tests against the original implementation and for debugging spectral boundary behavior.

Interior fourth-order centered first derivative:

$$(\partial_\xi f)_j = \frac{f_{j-2} - 8f_{j-1} + 8f_{j+1} - f_{j+2}}{12\Delta\xi} + \mathcal{O}(\Delta\xi^4). \quad (59)$$

Interior fourth derivative stencil:

$$(\partial_\xi^4 f)_j = \frac{f_{j-2} - 4f_{j-1} + 6f_j - 4f_{j+1} + f_{j+2}}{\Delta\xi^4} + \mathcal{O}(\Delta\xi^2). \quad (60)$$

The fallback parallel streaming operator uses sign-dependent upwinding. Boundary stencils follow GKW/Gyaradax. The implementation should store dimensionless stencil tables

$$S_{z,+}^{(1)}, \quad S_{z,-}^{(1)}, \quad S_{z,+}^{(4)}, \quad S_{z,-}^{(4)} \quad (61)$$

and select them by $\text{sign}(a_{\parallel,s})$. At a boundary, a stencil point outside the z -domain is resolved with `ixplus` or `ixminus`; if no connected mode exists, the point is treated with the chosen open-boundary rule.

For the fallback mirror term in $v_{\parallel}$, use the centered fourth-order derivative in the interior and zero-padding/no-flux style boundary behavior, matching GKW velocity-space conventions. Every derivative routine must have manufactured-solution tests in both spectral and fallback modes.

For production GKW parity with `ltrapping_arakawa`, the fallback also exposes a fused matrix-free `igh` operator. This operator does not apply Term I and Term IV as two independent derivatives. Instead it precomputes row-local source-shift weights $C_{\Delta v, \Delta z}$ with $\Delta v, \Delta z \in [-2, 2]$ and applies

$$(\mathcal{L}_{\text{igh}} f)_{a,b,j,k_x,k_y} = \sum_{\Delta v=-2}^2 \sum_{\Delta z=-2}^2 C_{\Delta v, \Delta z; a,b,j,k_x,k_y} f_{a+\Delta v, b, j+\Delta z, k'_x, k_y}. \quad (62)$$

The shifted k'_x is obtained from the same GKW mode-connectivity maps used by the upwind fallback. The coefficients include the Arakawa Hamiltonian stencil for $H = v_{\parallel}^2/2 + \mu B$ and the in-operator `disp_par` and `disp_vp` fourth-difference diffusion terms. This backend is a parity/debug path for the GKW finite-difference grid; the differentiable spectral target remains the default solver route.

9 Residual Assembly and Precomputation

9.1 Precomputed coefficient arrays

Before a time loop or eigenvalue solve, construct a precompute object

$$\mathcal{P} = \left\{ \begin{array}{l} J_{0,s}, \Gamma_{0,s}, F_{M,s}, \Xi_s, \omega_{d,s}, \\ a_{\parallel,s}, a_{\mu,s}, \\ D_z, D_{v_{\parallel}}, D_{\mu}, \quad T_z, T_{v_{\parallel}}, T_{\mu}, \\ \text{fallback stencil tables}, \\ \text{phi weights}, \quad \text{diagnostic weights} \end{array} \right\}. \quad (63)$$

Shapes for the adiabatic single-species path should be

$$\begin{aligned} f : & \quad (N_v, N_{\mu}, N_z, N_x, N_y), \\ \phi : & \quad (N_z, N_x, N_y), \\ J_0, F_M, \omega_d : & \quad (N_v, N_{\mu}, N_z, N_x, N_y), \\ a_{\parallel}, a_{\mu} : & \quad (N_v, N_{\mu}, N_z, 1, 1). \end{aligned} \quad (64)$$

Multi-species arrays add a leading species axis.

9.2 Public solver interfaces

The core interfaces should be pure functions:

$$\phi = \text{calculate_phi}(f, \mathcal{G}, p, \mathcal{P}), \quad (65)$$

$$r = \text{linear_residual}(f, \phi, \mathcal{G}, p, \mathcal{P}), \quad (66)$$

$$f^{n+1} = \text{rk4_step}(f^n, \mathcal{G}, p, \mathcal{P}). \quad (67)$$

No file I/O, Python-side mutation, random initialization, or diagnostic printing belongs inside these functions. This keeps the residual JIT-compatible and differentiable.

10 Time Integration and Linear Growth Rates

10.1 RK4

The baseline time integrator is explicit fourth-order Runge–Kutta:

$$k_1 = \mathcal{R}(f^n), \quad (68)$$

$$k_2 = \mathcal{R}(f^n + \tfrac{1}{2}\Delta t k_1), \quad (69)$$

$$k_3 = \mathcal{R}(f^n + \tfrac{1}{2}\Delta t k_2), \quad (70)$$

$$k_4 = \mathcal{R}(f^n + \Delta t k_3), \quad (71)$$

$$f^{n+1} = f^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (72)$$

Each RHS evaluation recomputes $\phi = \Phi(f)$. Multi-step runs should use a scan-style loop so that JAX can compile the full step block.

10.2 Timestep estimate

Use fixed Δt first. Add adaptive estimates after the fixed-step solver passes benchmarks. For the spectral backend, estimate the explicit stability limit using operator radii:

$$\rho_1 = \rho(a_{\parallel} D_z + a_{\mu} D_{\mu}), \quad \rho_{\text{diss}} = \rho(\mathcal{D}_z + \mathcal{D}_{v_{\parallel}} + \mathcal{D}_{\mu} + \mathcal{D}_{\perp}), \quad (73)$$

where $\rho(\cdot)$ is either an inexpensive power-iteration estimate or an upper bound from row sums on small grids. A practical RK4 estimate is

$$\Delta t_{\text{spec}} = c_{\text{safety}} \min \left(\frac{2.4}{\max(\rho_1, \epsilon)}, \frac{2.4}{\max(\rho_{\text{diss}}, \epsilon)} \right). \quad (74)$$

The finite-difference fallback may also expose the GKW-compatible linear estimate

$$\Delta t_{\text{lin}} = \frac{c_{\text{safety}}}{\max \left[40, \frac{t_1}{2.4}, \frac{t_4}{2.4}, \frac{t_{\text{field}}}{2.4} \right]}, \quad (75)$$

with

$$t_1 = \max \left(c_{z,1} \frac{\max |a_{\parallel}|}{\Delta z}, c_{v,1} \frac{\max |a_{\mu}|}{\Delta v} \right), \quad (76)$$

$$t_4 = \max \left(c_{z,4} \nu_z \frac{\max |a_{\parallel}|}{\Delta z}, c_{v,4} \nu_v \frac{\max |a_{\mu}|}{\Delta v} \right). \quad (77)$$

Here 2.4 is the RK4 stability-boundary factor used in GKW/Gyaradax notes. t_{field} is needed mainly for kinetic electrons and can be zero in the adiabatic baseline. The effective timestep is

$$\Delta t = \min(\Delta t_{\text{input}}, \Delta t_{\text{spec}}, \Delta t_{\text{lin}}, \Delta t_{\text{NL}}), \quad (78)$$

where Δt_{lin} is ignored unless the fallback backend is active, and Δt_{NL} is relevant only after the nonlinear ExB term is implemented.

10.3 Mode amplitude, growth rate, and frequency

Let $\mathcal{C}(k_y)$ be the connected k_x -chain containing $k_x = 0$. Define the per- k_y potential amplitude

$$A(k_y, t) = \left[\sum_z w_z \sum_{k_x \in \mathcal{C}(k_y)} |\phi(z, k_x, k_y, t)|^2 \right]^{1/2}. \quad (79)$$

Over a diagnostic window $[t_a, t_b]$,

$$\gamma(k_y) = \frac{\log A(k_y, t_b) - \log A(k_y, t_a)}{t_b - t_a}. \quad (80)$$

A robust real-frequency estimator is the phase rotation of the complex mode:

$$\omega(k_y) = -\frac{1}{t_b - t_a} \arg \left[\sum_z w_z \sum_{k_x \in \mathcal{C}(k_y)} \phi^*(z, k_x, k_y, t_a) \phi(z, k_x, k_y, t_b) \right]. \quad (81)$$

In linear initial-value runs, normalize each active k_y after each diagnostic window:

$$f_s(\dots, k_y) \leftarrow \frac{f_s(\dots, k_y)}{\max(A(k_y), \epsilon_A)}. \quad (82)$$

Store the accumulated normalization factor so diagnostics remain physically interpretable.

11 Fluxes and Quasilinear Objective

The immediate optimization objective is not nonlinear heat flux, but a differentiable linear proxy. For each k_y , define a mode-structure weighted perpendicular scale

$$\langle k_{\perp}^2 \rangle_{k_y} = \frac{\sum_z w_z \sum_{k_x \in \mathcal{C}(k_y)} k_{\perp}^2(z, k_x, k_y) |\phi(z, k_x, k_y)|^2}{\sum_z w_z \sum_{k_x \in \mathcal{C}(k_y)} |\phi(z, k_x, k_y)|^2 + \epsilon_A}. \quad (83)$$

The first quasilinear proxy is

$$Q_{\text{QL}} = \sum_{k_y > 0} W(k_y) \frac{\max(\gamma(k_y), 0)}{\langle k_{\perp}^2 \rangle_{k_y} + \epsilon_k}. \quad (84)$$

For smooth reverse-mode differentiation, replace $\max(\gamma, 0)$ by a softplus if needed:

$$\max(\gamma, 0) \approx \tau \log(1 + e^{\gamma/\tau}). \quad (85)$$

This objective matches the spirit of direct microstability optimization while remaining cheap enough for early design-loop tests.

12 Differentiability Contract

The following quantities should be differentiable:

- continuous geometry arrays $B, F, G, E_y, D_x, D_y, g^{xx}, g^{xy}, g^{yy}$;
- profile gradients ω_n, ω_T ;
- species parameters where physically meaningful;
- scalar geometry knobs in analytic tests, e.g. $q, \hat{s}, \epsilon$;
- objective functions $A, \gamma, \omega, Q_{QL}$ through a fixed number of time steps.

The following quantities are static/non-differentiable:

- grid sizes;
- integer mode labels and connectivity maps;
- branch choices for topology;
- file I/O and equilibrium loading;
- random initialization keys.

For stellarator optimization, a differentiable equilibrium code may provide geometry arrays and their sensitivities. The gyrokinetic solver should not require differentiating through an integer remeshing or topology change. The implementation enforces this boundary with a versioned topology contract that fingerprints velocity and parallel nodes and backends, perpendicular mode sets, twist-and-shift connectivity, and provider field-line metadata. A changed fingerprint invalidates compiled objectives, checkpoints, and gradients. In the present implementation, imported DESC-style arrays are accepted as a `desc` geometry model in the single-surface objective; the direct DESC object adapter is optional and remains outside the solver-core dependency chain.

13 Current Implementation and Validation Status

The implementation now covers the linear electrostatic architecture specified above. The core package in `src/jax_fluxtube_gk/` contains PyTree parameter objects, Fourier grids, spectral and finite-difference derivative backends, analytic and imported stellarator geometry adapters, quasineutrality solvers, term-level linear RHS assembly, RK4 time advance, growth/frequency diagnostics, stable design objectives, gradient audits, robust multi-sample aggregation, fixed-topology guards, and schema-versioned optimization checkpoints. The production physics target remains linear electrostatic gyrokinetics with adiabatic electrons; nonlinear turbulence and extended physics are deferred to [Appendix A](#).

13.1 Validation guardrails

The current regression suite is intentionally layered. Algebraic tests cover broadcasting, PyTree behavior, derivative operators, field solves, term linearity, and differentiability on reduced grids. Code-to-code and benchmark fixtures then check the pieces most likely to hide convention errors:

- the true Rosenbluth–Hinton late-plateau metric passes against the GKW/Gyaradax value 0.0711, with observed value 0.070413;

- the Cyclone selected- k_y scalar growth gate passes after matching GKW’s internal wave-number convention;
- GKW selected-mode RHS/action traces, imported-state one-window replays, initial-state normalization, and row-normalized `parallel_phi.dat` profile checks pass at their current tolerances;
- eik geometry checks pass for DESC fixtures, solver export, eik import, and independent VMEC/GIST sources;
- reduced stellarator scan and DESC optimization examples execute through fixed-topology differentiable objectives, but are not production claims.

These passing gates are guardrails, not the end state. Two confidence gaps are kept visible: full selected-mode GKW state-history parity remains open, and the multi-time GKW velocity-slice error grows from 3.99×10^{-3} at step 20 to 3.67×10^{-2} by step 800. They do not currently block the W7-X external-reference path, but they remain useful warnings against overclaiming.

13.2 One-command stellarator scans

The reduced imported-geometry scan is executable with

```
uv run python examples/run_stellarator_linear_scan.py
```

By default it loads `fixtures/desc_geometry_dshape_rho05_alpha0.npz`, runs a short fixed-topology linear scan, and writes geometry audits, growth/frequency tables, mode structures, convergence metadata, and a quasilinear proxy. The same command can read DESC paths, eik tables, and stella `.geometry` files. Before solving, it applies the shared geometry-preflight contract: finite fields, positive B , positive metric diagonals, representative nonnegative $k_\perp^2$, eik-export compatibility, and an optional finite-difference mirror-force check when the source data supports it. The separate mode-boundary contract checks zonal identity maps, nonzonal open chain ends, reciprocal linked shifts, and `ikxspace` spacing.

A real-geometry reduced W7-X fixture is stored in `fixtures/w7x_itg_reduced_benchmark/`. It uses a W7-X eik geometry and adiabatic-electron ITG input as provenance. Its scalar outputs are finite regression artifacts, not an external growth-rate claim; the fixture is deliberately guarded by production-readiness ledgers.

13.3 Matched W7-X stella reference

The preferred external W7-X reference is stella, a CPU continuum gyrokinetic code that can run the matched case locally. Acceptance uses `torflux` = 0.64, α = 0, adiabatic electrons, k_x = 0, k_y = 0.3, N_z = 256, $N_{v\parallel}$ = 32, N_μ = 8, Δt = 0.1, and t = 500. Generated stella input, geometry, state, and output remain external scratch data rather than canonical repository assets. The lower k_y = 0.1, 0.2 branches are diagnostic because their stella omega windows fail the convergence test even at t = 1000.

The solver imports the external stella `.geometry` table only for this independent validation path. With the source-matched analytic initial perturbation, the converged branch gives

$$(|\Delta\gamma|, |\Delta\omega_r|, \epsilon_\phi) = (3.20 \times 10^{-4}, 2.18 \times 10^{-3}, 1.08 \times 10^{-2}),$$

where ϵ_ϕ is the phase-aligned complex profile error. All three values pass the 2×10^{-2} gate.

13.4 Resolved W7-X timestep discriminator

Source instrumentation established the production ordering, which differs from a generic RK4 Strang split. Each stella step consists of three SSP RK3 explicit stages, one full sign-dependent cubic semi-Lagrangian mirror characteristic, and one full implicit parallel-streaming and quasineutrality response. On an exact native stella input, the JAX replay errors after RK3, after mirror advance, in the final distribution, and in potential are respectively

$$7.19 \times 10^{-6}, \quad 1.34 \times 10^{-4}, \quad 2.17 \times 10^{-5}, \quad 1.34 \times 10^{-4}.$$

This closes the timestep discriminator independently of long-time eigenmode selection.

The earlier long-time profile mismatch came from comparing different initial mode mixtures. Stella initializes a localized Maxwellian perturbation,

$$g_0 = A(1 + i) \exp[-(2\pi z)^2] \exp[-v_{\parallel}^2 - 2\mu B(z)],$$

whereas the generic scan used a dense oscillatory state. The `stella_maxwellian` option implements this expression analytically; no W7-X distribution is stored. This distinction matters at finite time for a nonnormal spectrum with slowly separating branches.

13.5 Nonlinear turbulence acceptance status

The electrostatic nonlinear-turbulence acceptance campaign (Appendix A) is implemented but not scientifically closed; this is a compute-evidence gap, not a missing operator. Three of its four gates have passing rungs:

- a stationary phase-space resolution ladder, $16 \times 16 \times 8 \rightarrow 20 \times 20 \times 10$ at fixed Fourier/physics contract, changes the mean local heat-flux proxy by 3.88%, below the 15% gate;
- a three-lineage ensemble with distinct initialization roots (seed, amplitude, zonal fraction) is stationary with maximum relative deviation 2.04% from the ensemble mean, below the 15% gate;
- the bandwidth-preserving domain-convergence ladder is *not* converged: the stationary $65 \times 33 \rightarrow 129 \times 65$, $k_{y,\min} = 0.0125 \rightarrow 0.00625$ finest-pair change is 25.65%, above the 15% gate, so a further 257×129 , $k_{y,\min} = 0.003125$ bandwidth-preserving rung is required;
- independent nonlinear parity against a revision-pinned external reference gyrokinetic code remains deferred: it needs a CUDA-capable host, which was not available for this campaign.

The required next domain rung was attempted on a CPU-only development workstation and ran for approximately 19.5 hours without a solver or memory error, but did not reach its checkpoint/report phase before being deliberately terminated; no partial artifact was produced or retained. Both remaining gates need external compute this project does not currently have access to (a CPU cluster or GPU/CUDA host), not further algorithmic development. Unrestricted equilibrium-shape optimization (Appendix A) is explicitly gated behind all four nonlinear-acceptance items and is not attempted before they pass.

13.6 Optimization status

The reduced fixed-topology optimization milestone is complete. The public objective contract composes selected, hard-maximum, or smooth-maximum growth, real-frequency targets, phase-invariant complex mode-structure penalties, and quasilinear proxies. Reverse-mode derivatives are checked against central differences through geometry arrays and time advance, with explicit

diagnostics for near-degenerate k_y branches. Fixed scans over ρ , α , and k_y support weighted-mean, hard-worst-case, and differentiable soft-worst-case reduction. Checkpoints retain the complete sample tensor, objective policy, topology fingerprints, provider provenance, revisions, command, seed, parameters, and iteration history.

A reduced W7-X loop loads the installed VMEC++ named configuration, performs fresh center/plus/minus equilibrium solves for a boundary harmonic, converts each result in memory, verifies fixed topology, and takes an outer finite-difference step. No equilibrium artifact is stored in the repository. This demonstrates the real MHD provider boundary but does not differentiate through VMEC++, claim unrestricted equilibrium-shape optimization, or predict nonlinear turbulent heat flux.

13.7 Optimization results

This section reports two distinct classes of executed optimization evidence. The first is a genuine multi-step, end-to-end differentiable optimization on analytic circular geometry: every iteration differentiates through the full residual, field solve, and time advance via `jax.grad`. The second is a real MHD-coupled boundary-harmonic scan through the installed DESC, GVEC, and VMEC++ providers: a fresh equilibrium is re-solved at each of three boundary-scale points and the resulting geometry is fed through the unchanged gyrokinetic objective. The two are not the same claim. The first exercises the differentiable optimizer on a fixed, cheap analytic geometry; the second exercises the live equilibrium/geometry/objective pipeline but only as a three-point central-difference sensitivity, not a multi-step optimizer, and is not a claim of converged shape optimization. Section 13 already documents that end-to-end automatic differentiation *through* a live equilibrium solve is a separate, unimplemented milestone; both result classes below are consistent with that boundary.

13.7.1 Differentiable knob optimization

`examples/generate_optimization_figures.py` runs three 1000-iteration gradient-descent cases (A , B , C) that minimize $\frac{1}{2}(\gamma_{\text{obj}} - \gamma_{\text{target}})^2$ with $\gamma_{\text{target}} = 0$ (drive to marginal stability) by differentiating through q , $\hat{s}$, R/L_n , R/L_T , and the other circular-geometry knobs in `OptimizationKnobs`. Case A targets a single selected k_y branch on a two-mode grid; case B targets a single selected branch on a three-mode grid with a different starting equilibrium; case C targets the differentiable soft/hard maximum over the nonzonal branches (`scalar_mode="max_nonzonal"`) rather than one fixed mode index. All three converge:

Case	Initial obj. error	Final obj. error	q : initial→final	$\hat{s}$: initial→final
A	6.47×10^{-3}	3.78×10^{-8}	$1.250 \rightarrow 1.261$	$0.450 \rightarrow 0.448$
B	7.85×10^{-2}	$< 10^{-12}$	$1.350 \rightarrow 1.371$	$0.650 \rightarrow 0.661$
C	6.72×10^{-2}	$< 10^{-12}$	$1.450 \rightarrow 1.468$	$0.750 \rightarrow 0.759$

Table 2: Reduced circular-geometry gradient-descent cases: absolute objective error and geometry-knob drift from 1000 `jax.value_and_grad` steps at learning rate 10^{-3} . Case A’s selected- k_y growth rate moves from 1.138×10^{-1} to 2.748×10^{-4} ; case C’s differentiable non-zonal maximum reaches its zero target while the specific selected branch (index 1, not the arg-max) settles at a distinct, more stable value -3.166×10^{-1} , showing the soft-maximum objective correctly tracks the least-stable branch rather than a fixed mode index.

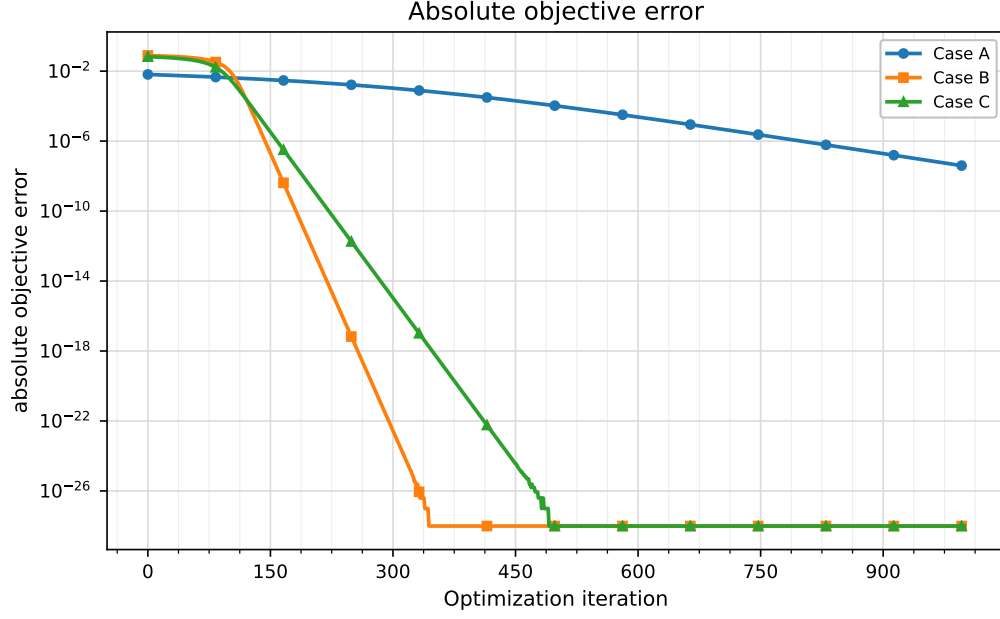


Figure 1: Absolute objective error versus iteration for cases A–C (log scale). All three cases converge monotonically to within numerical noise of the marginal-stability target.

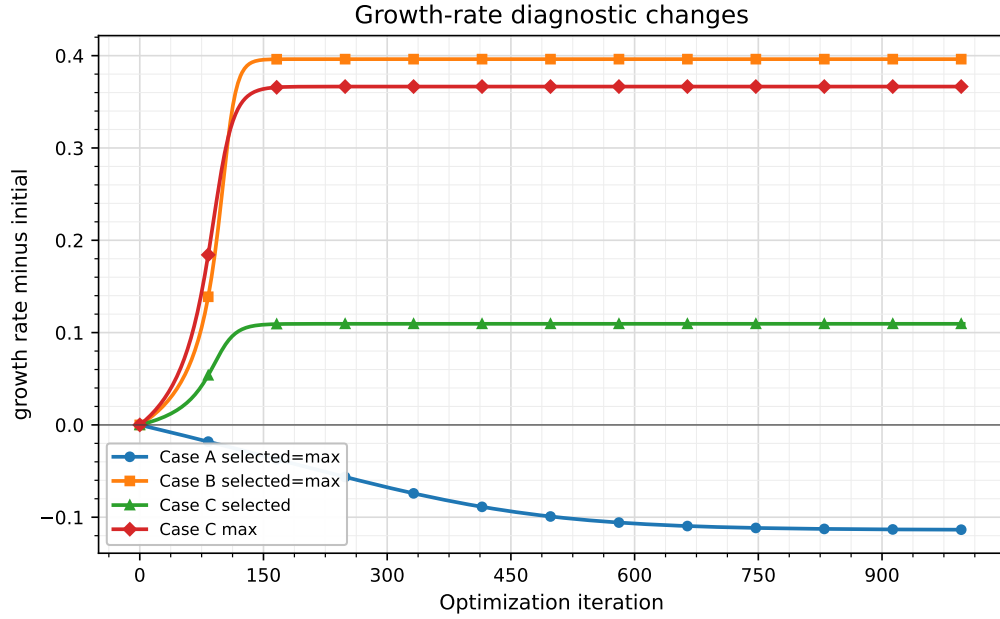


Figure 2: Selected-mode and nonzonal-maximum growth-rate change relative to each case's initial value. Case C shows the distinct selected-branch and max-nonzonal-branch trajectories referenced in Table 2.

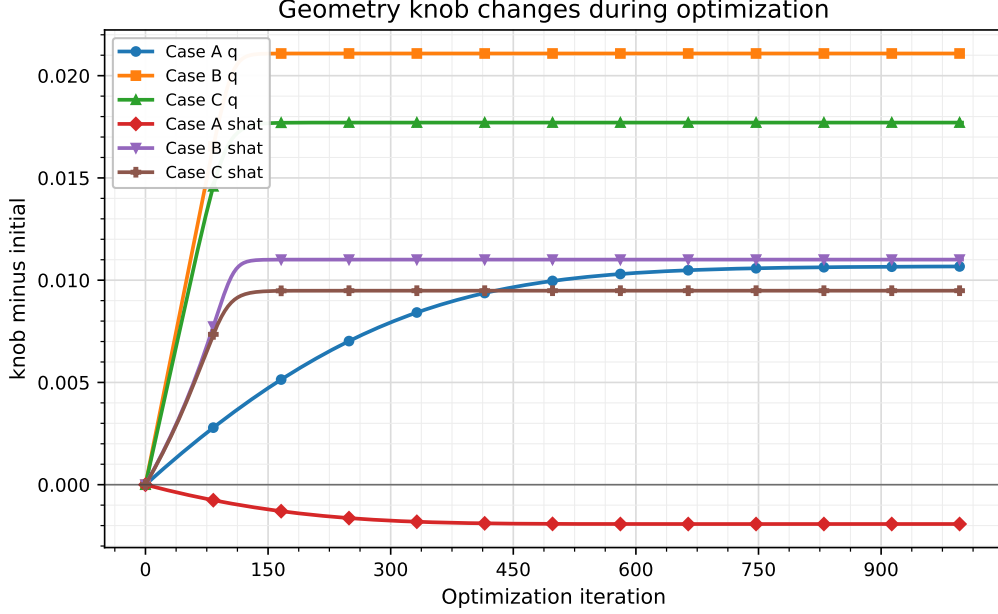


Figure 3: Safety-factor q and magnetic-shear $\hat{s}$ drift from their initial values under the gradient-descent update. Knob changes stay small (a few percent) over the full trajectory, consistent with driving toward a nearby marginal-stability point rather than an unconstrained large excursion.

13.7.2 Live-provider design-loop results

Each of the three geometry-optimization notebooks (`examples/desc_geometry_optimization.ipynb`, `examples/gvec_geometry_optimization.ipynb`, `examples/vmecpp_geometry_optimization.ipynb`) was executed end to end against its installed live provider: DESC’s named DSHAPE configuration, VMEC++’s named **w7x-standard** configuration, and GVEC’s tracked VMEC-initialized W7-X parameterization. Each notebook scales one boundary Fourier harmonic by $\{1 - \delta, 1, 1 + \delta\}$, re-solves the equilibrium at every scale, reconstructs the flux-tube geometry, and re-evaluates the unchanged reduced gyrokinetic objective, verifying fixed optimization topology between solves with `assert_fixed_optimization_topology`.

Provider	Harmonic	Scale δ	Growth rate	Frequency	Central-diff. sensitivity
DESC	$R_{2,0}$	0.01	0.15295 $\rightarrow$ 0.15287	-0.0928 $\rightarrow$ -0.0907	-4.14×10^{-3}
GVEC	$X_{1,1}^{\cos}$	0.005	0.27422 $\rightarrow$ 0.27414	-0.0498 $\rightarrow$ -0.0498	-7.63×10^{-3}
VMEC++	$(m, n) = (1, 0)$	0.005	-0.21841 $\rightarrow$ -0.22166	0.1155 $\rightarrow$ 0.1160	-3.26×10^{-1}

Table 3: Live-provider boundary-harmonic scans. “Growth rate”/“Frequency” span the three evaluated scale points $1 - \delta$ to $1 + \delta$; the central-difference sensitivity is $(\text{obj.}(1 + \delta) - \text{obj.}(1 - \delta)) / (2\delta)$ of the scalar reduced-growth objective. GVEC’s minimizer was run at a bounded `MaxIter=100` and did not reach its force-tolerance criterion ($|\text{force}| \approx 2.3 \times 10^{-3}$ against a 10^{-8} target) at any of the three scales, so its geometry is a diagnostic near-equilibrium, not a converged one – consistent with each notebook’s own framing of this scan as an interface/sensitivity demonstration rather than an accepted design point.

None of the three scans differentiates through the equilibrium solve itself; each is an outer-loop, finite-difference sensitivity through independently re-solved equilibria, matching the “Outer loop (available now)” path described in the notebooks rather than the deferred end-to-end automatic-differentiation path.

Extended outer-loop hill-climb. Beyond the single 3-point sensitivity in Table 3, each notebook was extended into a longer outer-loop hill-climb targeting 100 iterations: the same central-difference sensitivity steps the boundary scale every iteration, and the loop fails closed – stopping early but keeping every completed iteration – on a non-finite objective, a raised solver exception, or a wall-clock budget. DESC completed 46 iterations before a boundary-harmonic scale of 1.0037 produced a non-finite objective from a failed equilibrium solve; GVEC completed 48 iterations before its 900-second budget; VMEC++, whose equilibrium solve is roughly two orders of magnitude more expensive per evaluation than DESC or GVEC at this resolution, completed 15 iterations within an extended 1800-second budget using a finer 3-iteration frame stride (stopping after 1877s against that budget). None of these stops is a bug: each is the fail-closed loop behaving exactly as designed when a real equilibrium solver becomes unreliable or a shared compute budget is exhausted, and each is reported rather than hidden.

Figures 4–8 show the real solved 3D boundary surface at the recorded frame stride, reconstructed directly from each provider’s own boundary representation (DESC’s `FourierRZToroidalSurface` via `surface.compute`, GVEC’s Cartesian `state.evaluate("pos", ...)` at $\rho = 1$, and VMEC++’s `rbc/zbs` Fourier reconstruction) rather than a schematic. Figures 5–9 show the corresponding objective trajectories: DESC and GVEC both decrease monotonically toward the marginal-stability target as the boundary scale increases; VMEC++’s objective also decreases monotonically, from -0.22007 to -0.23109 , as its boundary scale moves from 1.0 to 1.0394 over the same 15 iterations – a visibly larger single-iteration step than DESC or GVEC, consistent with its order-of-magnitude larger central-difference sensitivity in Table 3.

DESC boundary evolution across the outer design loop (every 25 iterations)

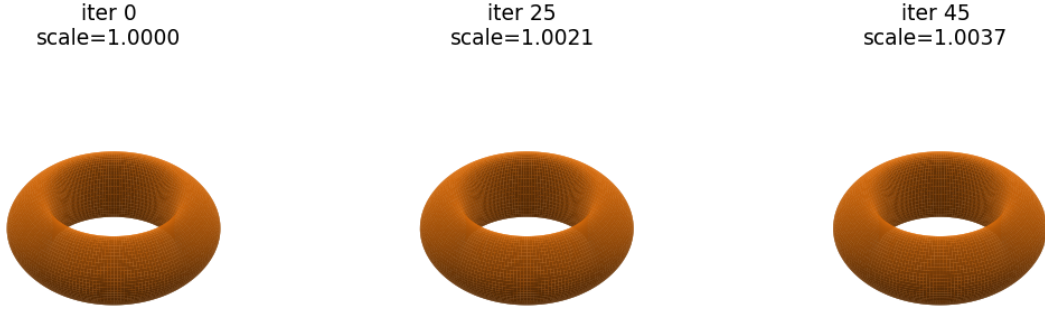


Figure 4: DESC $R_{2,0}$ boundary evolution across 46 completed outer-loop iterations, shown every 25 iterations. Each surface is evaluated directly from the solved equilibrium’s boundary via `surface.compute`.

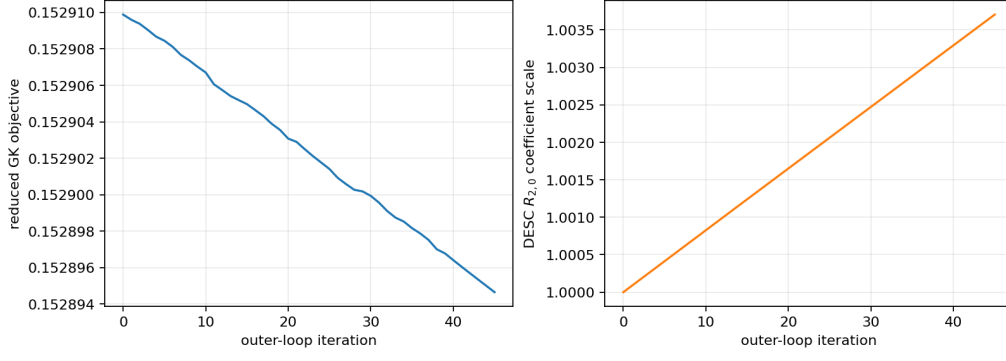


Figure 5: DESC reduced GK objective and $R_{2,0}$ coefficient scale across the same 46-iteration trajectory.

GVEC W7-X boundary evolution across the outer design loop (every 25 iterations)

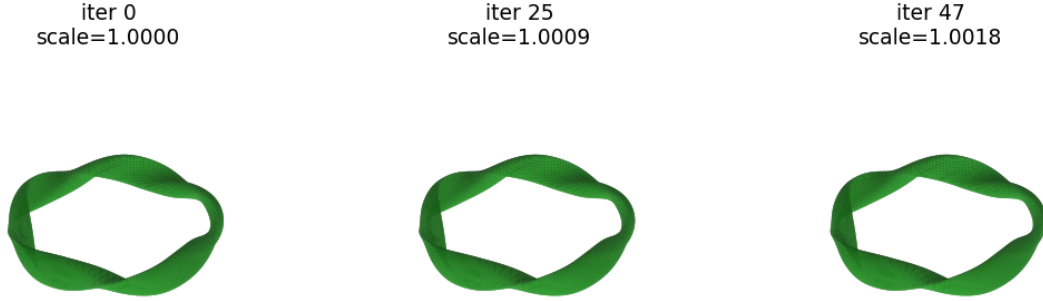


Figure 6: GVEC $X_{1,1}^{\cos}$ boundary evolution across 48 completed outer-loop iterations, shown every 25 iterations. Each surface is the real Cartesian position of the solved GVEC state at $\rho = 1$.

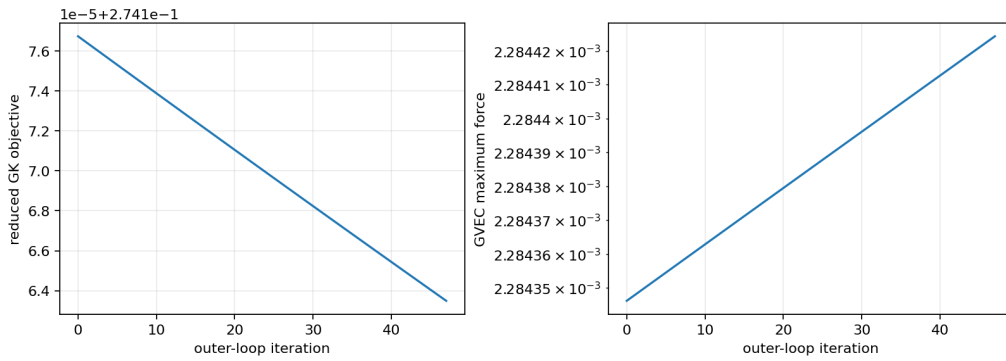


Figure 7: GVEC reduced GK objective and maximum force residual across the same 48-iteration trajectory. The force residual stays flat and bounded away from GVEC's own convergence tolerance throughout, consistent with the bounded-`MaxIter` diagnostic framing used everywhere else in this section.

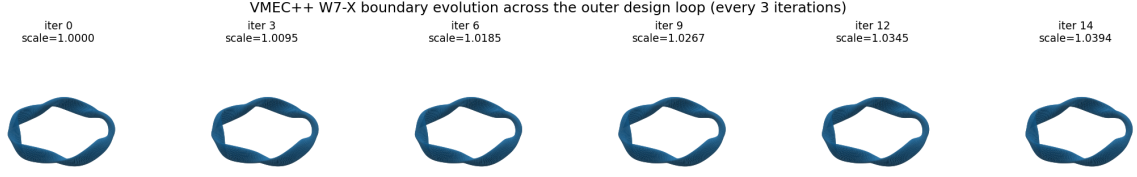


Figure 8: VMEC++ paired $(m, n) = (1, 0)$ boundary evolution across 15 completed outer-loop iterations, shown every 3 iterations because of VMEC++’s substantially higher per-solve cost at this resolution. Each surface is reconstructed directly from the scaled `rbc/zbs` Fourier coefficients VMEC++ solved for.

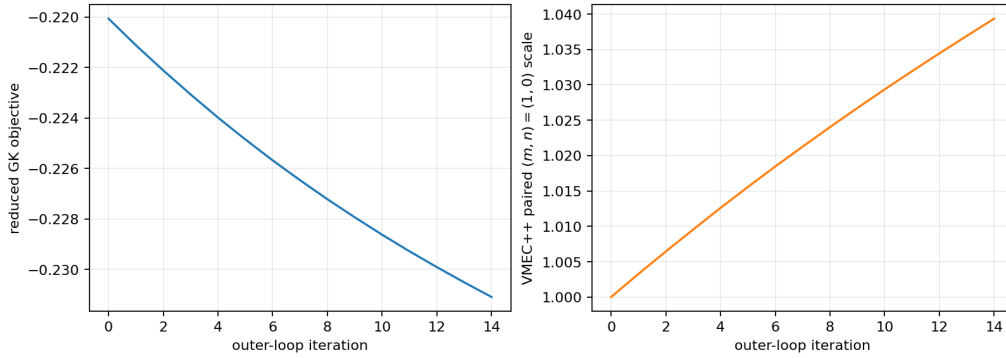


Figure 9: VMEC++ reduced GK objective and paired $(m, n) = (1, 0)$ scale across the same 15-iteration trajectory.

13.8 Production criteria

The trusted linear W7-X branch now satisfies:

1. weighted stella-vs-solver $k_y = 0.3$ term and stage-array parity;
2. a passing converged W7-X growth, frequency, and mode-profile gate;
3. source-matched field-line, 256-point parallel, 32-by-8 velocity, timestep, split-ordering, initial-state, and growth-window controls;
4. guarded CPU timing of the validated advance after parity passes.

Production full-shape design optimization additionally requires a differentiable equilibrium provider (or a validated sensitivity interface) and physics beyond the present reduced linear objective. A broader low- k_y spectral claim additionally requires converged independent reference windows.

14 Testing and Benchmark Contract

Every public function in `src/` should have focused tests. For a narrow change, update the smallest relevant unit or fixture test; for a change that alters shared physics, geometry, time advancement, or validation contracts, run the corresponding focused gate plus the broader suite.

14.1 Required unit-test coverage

Component	Required tests
Parameter PyTrees	flatten/unflatten, static/dynamic fields, JIT behavior.
Grids and topology	shapes, centering, quadrature weights, Parseval weights, mode labels, linked/open boundary maps.
Derivative backends	manufactured derivative checks, finite-difference parity, filter behavior.
Geometry	finite fields, $B > 0$, $k_{\perp}^2 \geq 0$, eik/DESC/stella adapter contracts, differentiability of continuous arrays.
Physics primitives	Bessel/FLR limits, Maxwellian and drive factors, mirror/drift/streaming coefficients, broadcasting, AD checks.
Field solve	zero-input behavior, zonal and nonzonal paths, quasineutrality residuals.
RHS terms	zero-input behavior, linearity, isolated term formulas, full residual reconstruction, JIT compatibility.
Time advance	RK4 order, fixed-step scans, growth/frequency recovery, phase-unwrapped frequency on long windows.
Objectives	finite values, gradients, finite-difference checks on reduced fixed-topology problems.
Validation scripts	committed fixture schemas, status JSON contracts, failure states, and command-line smoke paths.

14.2 Benchmark ladder

The benchmark order should remain:

1. manufactured derivatives, field solves, and scalar ODE tests;
2. Gyrodax circular and s-alpha parity;
3. GKW simple/Cyclone fixtures and selected-mode RHS/action traces;
4. Rosenbluth-Hinton late plateau;
5. Cyclone selected- k_y scalar and mode-structure diagnostics;
6. DESC/stella eik and geometry contracts;
7. matched W7-X stella growth, frequency, and complex $\phi(z)$ parity;
8. production-control W7-X convergence and CPU timing;
9. only then DESC optimization campaigns and nonlinear extensions.

14.3 First trusted solver milestone

The first trusted solver milestone is complete when $\Phi(f)$, $\mathcal{R}(f)$, RK4 stepping, γ , ω , and Q_{QL} are implemented, the reduced RH/Cyclone/GKW/DESC/eik guardrails pass, a matched W7-X stella mode-structure gate passes, production-control convergence and CPU timing are recorded, and `STATUS.md` states the remaining caveats without using reduced examples as production evidence.

A Nonlinear and Extended Physics Roadmap

The nonlinear ExB term retained for later is

$$\mathcal{T}_{III,s} = -\{J_{0,s}\phi, f_s\}, \quad \{a, b\} = \partial_x a \partial_y b - \partial_y a \partial_x b. \quad (86)$$

The implementation should be pseudospectral:

1. multiply by ik_x and ik_y in Fourier space;
2. inverse FFT to a dealiased real grid;
3. compute the bracket in real space;
4. FFT back and truncate to the retained spectral grid.

The implemented nonlinear foundation follows this construction for centered k_x modes and the nonnegative- k_y Hermitian half spectrum. It also evaluates an advective timestep bound from the padded-grid ExB velocity. The host-controlled nonlinear driver chooses the minimum of this instantaneous bound and the coupled linear bound for each accepted RK4 step. Since timestep accept decisions are nonsmooth, differentiable trajectories retain the fixed-step driver. Production nonlinear readiness additionally requires statistically stationary heat flux, resolution and box-size convergence, and independent-code parity.

For a collocation-space state, the implemented electrostatic heat response is

$$\mathcal{Q}_{s,\mathbf{k}}(z) = \int d^3v J_{0,s} T_s \left(\mathcal{E}_s - \frac{3}{2} \right) f_{s,\mathbf{k}}, \quad (87)$$

and its radial spectral contribution is proportional to $k_y \text{Im}[\phi_{\mathbf{k}}^* \mathcal{Q}_{s,\mathbf{k}}]$. The velocity integral uses the same discrete measure as the field solve. This defines the evolved-state diagnostic but does not replace the stationarity, resolution, domain-size, and independent-code gates.

Other extensions should be added only after the linear electrostatic baseline is validated:

- kinetic electrons via (45);
- collisions from `collisionop.F90`;
- electromagnetic $A_{\parallel}$ and $B_{\parallel}$;
- toroidal rotation and ExB shear from `rotation.F90`;
- nonlinear heat-flux objectives.

As an intermediate collision model, the collocation solver may use a species-local linearized BGK operator

$$C_s[f_s] = -\nu_s (f_s - \mathcal{P}_s f_s), \quad (88)$$

where the discrete projection $\mathcal{P}_s$ spans $F_{M,s}$, $v_{\parallel} F_{M,s}$, and $\mathcal{E}_s F_{M,s}$ in the same quadrature used by the field solve. This construction conserves the discrete density, parallel-momentum, and energy moments species by species. It is a differentiable model-collision foundation, not a replacement for a validated Landau/Fokker–Planck operator with inter-species exchange.

The finite-difference Fokker–Planck development path also retains the ordered test-particle contributions $C_{ab}[f_a]$. Its experimental pairwise completion groups $C_{ab}[f_a]$ and $C_{ba}[f_b]$, then applies a discrete projection for the two density constraints and the pair’s combined physical parallel-momentum and energy constraints. Consequently, the directed contributions satisfy

$$\int C_{ab}^{\text{pair}} d^3v = 0, \quad \sum_{s \in \{a,b\}} \int \left(\frac{p_{\parallel,s}}{E_s} \right) C_{ss}^{\text{pair}} d^3v = 0. \quad (89)$$

This creates reciprocal cross-species data flow and a pairwise conservation contract. It is not the Laguerre–Legendre field-particle response used by stella and must not be described as a validated Landau operator until its coefficients and action pass an independent executable parity gate.

The validated software form for a future Laguerre–Legendre completion is the low-rank pair/component contraction

$$\Psi_{ab,c}(z, \mathbf{k}) = \sum_{v_{\parallel}, \mu} D_{ab,c}(v_{\parallel}, \mu, z, \mathbf{k}) f_b, \quad C_a^{\text{fp}} = \sum_{b,c} R_{ab,c} \Psi_{ab,c}, \quad (90)$$

where $c = (l, m, j)$, D includes the discrete velocity measure and all driver normalizations, and R is the target-space response. The JAX implementation accepts these tensors without depending on stella. Native stella traces verify every stage of this construction: the driver D , the response R (built independently from the physical velocity and magnetic grids, species mass ratios, pair collision frequencies, gyroaverages, and Δ_j), and their low-rank contraction reproduce tens of thousands of native reference coefficients and the native solved- Ψ action at relative L_2 errors between 10^{-11} and 10^{-16} , independent of the equilibrium state.

The remaining piece, the differential test-particle matrix C_{tp} , is independently constructed rather than read from stella: the Gyradax-matched nine-point stencil; the mu- and parallel-velocity finite-difference blocks, including their interior and boundary closures on uniform and nonuniform grids; the mode-dependent gyro diagonal; and the underlying analytic collision-frequency layer (speed, Maxwellian, parallel diffusion, deflection, mixed diffusion). Each piece matches the corresponding native stella matrices and, for the four ordered species channels, the all-channel matrix and final 2,808-value phase-space state, at relative L_2 errors between 10^{-14} and 10^{-17} . A differentiable Woodbury solve combines $I - \Delta t C_{\text{tp}}$ with D and R in stella’s implicit ordering and agrees with a directly materialized dense backward-Euler system at roundoff.

The resulting production implicit collision backend – selectable as a split step alongside the linear residual – is therefore validated end to end against native stella at near-roundoff precision, closing the collision implementation item. Together with the independent electromagnetic parity gates in Sec. 13, production collisional and linear electromagnetic physics are implemented and validated for the accepted scope; nonlinear-turbulence and unrestricted design-optimization evidence remain the open items tracked in Sec. 13.5.

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